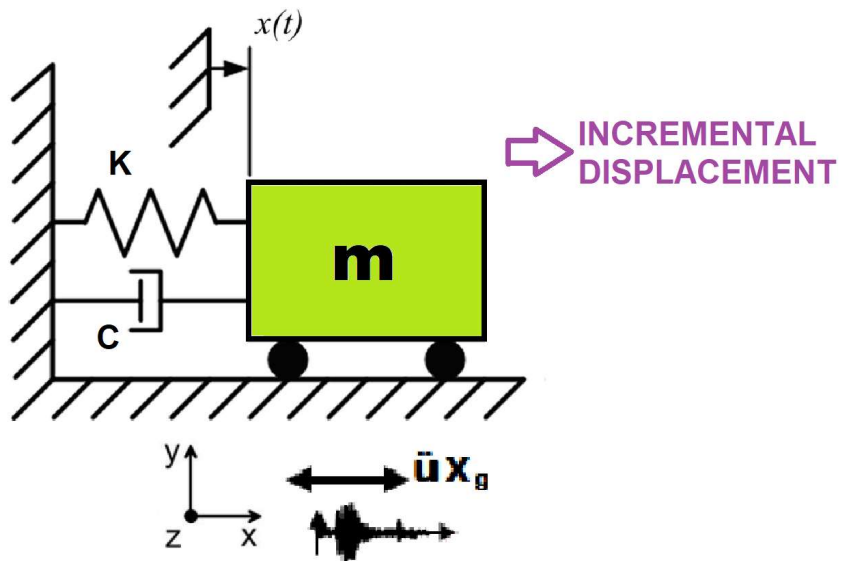


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**DETERMINATION OF OPTIMUM
STRUCTURAL DUCTILITY RATIO
FROM ENERGY DISSIPATION
CAPACITY INDEX USING
FINITE-DIFFERENCE NEWTON
ITERATION AND OPENSEES VIA 20
SEISMIC GROUND MOTIONS**

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

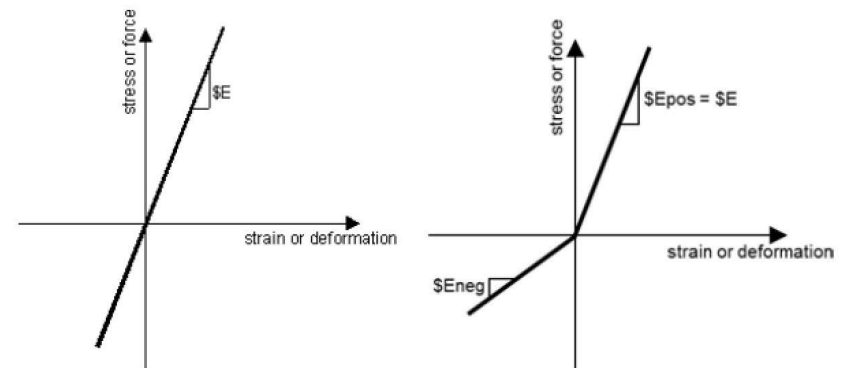
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

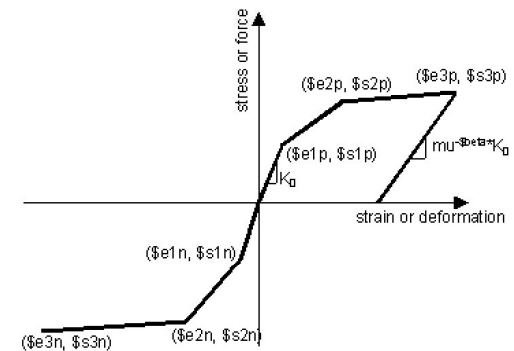
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

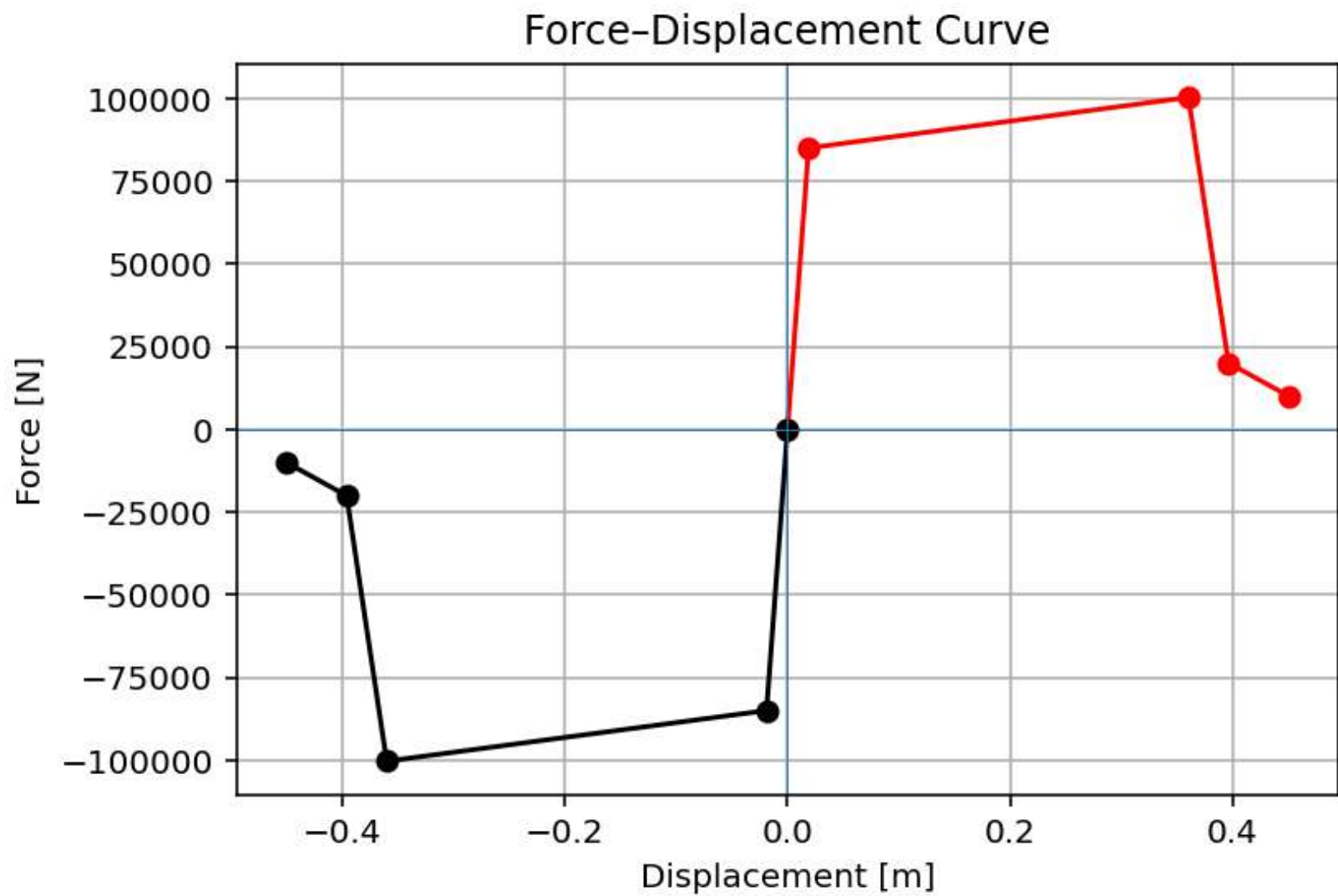
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



C:\Users\ DELL\Desktop\OPENSEES_FILES\HEIGHT_ANALY...C_SDOF+\DUCT_OPTIMIZATION_DECI_20_SEISMIC_SDOF.py

DUCT_OPTIMIZATION_...20_SEISMIC_SDOF.py X

```

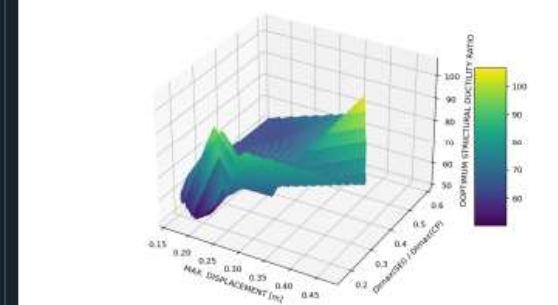
1 #####
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4 # FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES VIA 20 SEISMIC GROUND MOTIONS
5 # -----
6 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPENSEES FRAMEWC
7 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS
8 # -----
9 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
10 # -----
11 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$ 
12 # -----
13 #  $P(t) = P_0 \sin(\omega t)$ 
14 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
15 # -----
16 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
17 # OF ENERGY DISSIPATION CAPACITY INDEX
18 # -----
19 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
20 # EMAIL: salar.d.ghashghaei@gmail.com
21 #####
22 """
23 Nonlinear Seismic Performance Assessment of a SDOF:
24 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake Lc
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26 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
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28 (elastic-perfectly plastic or hysteretic steel with strain hardening)
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30 and large displacements—essential for collapse assessment.
31
32 Eight analysis protocols are implemented:
33 (1) [PERIOD] : Structural Period
34 (2) [STATIC] : Gravity Load analysis establishing dead load state

```

...NDEX\11_SDOF_8_ANA\DUCT_OPTIMIZATION_DECI_20_SEISMIC_SDOF+

26 %

3D CONTOUR: MAX. DISPLACEMENT [m] - Dmax(SEI) / Dmax(CPI) - OPTIMUM STRUCTURAL DUCTILITY RATIO



OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)

Assume that a single-degree-of-freedom structure is subjected to nonlinear dynamic analysis under twenty different ground motion records, and, by defining the amount of Energy Dissipation Capacity Index as the objective, we aim to optimize the ductility of the structure using the Newton–Raphson method. Therefore, with twenty different ground motion records, we calculate different ductility ratios, and finally, we evaluate the results.

FileEditSearchSourceRunDebugConsolesProjectsToolsViewHelp

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...C_SDOF+\DUCT_OPTIMIZATION_DECI_20_SEISMIC_SDOF.py

DUCT_OPTIMIZATION_...20_SEISMIC_SDOF.py * X

```
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11 # ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
12 #-----
13 # P(t) = P0 sin(wt)
14 # P(t) = P0 exp(-0.05wt) sin(wt)
15 #-----
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```

Console 1/A X

```
Time: 19.9900, Displacement: 0.1122 mm, Reaction: 124.70 N

+-----+
| lambda | omega | period | frequency |
+-----+
| 7.129e-01 | 0.8443 | 7.4418 | 0.1344 |
+-----+

Time: 20.0000, Displacement: 0.1122 mm, Reaction: 125.75 N

SEISMIC ANALYSIS DONE.

Dissipated Energy from Earthquake= 168051.84 N.m
Dissipated Energy from Cyclic Displacement= 840276.69 N.m

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

ZONE 2: MINOR DAMAGE

Fmax: -0.0004159685131490676
DF: -0.4163895489490699
DX: -9.906465844616561e-07
IT: 102 - RESIDUAL: 9.906465844616561e-07 X: 49.946969944869544

Seismic: 60
Optimum X (STRU. DUCTILITY RATIO): 49.9470
Iteration Counts: 102
Convergence Residual: 9.9064658446e-07
```

IPython Console Plots Variable Explorer Help Debugger History Files

File Edit Search Source Run Debug Consoles Projects Tools View Help

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DUCT_OPTIMIZATION_...20_SEISMIC_SDOF.py * X

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Console 1/A X

```
Time: 19.9900, Displacement: 0.1122 mm, Reaction: 124.70 N

+-----+
| lambda | omega | period | frequency |
+-----+
| 7.129e-01 | 0.8443 | 7.4418 | 0.1344 |
+-----+

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DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

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IT: 102 - RESIDUAL: 9.906465844616561e-07 X: 49.946969944869544

Seismic: 60
Optimum X (STRU. DUCTILITY RATIO): 49.9470
Iteration Counts: 102
Convergence Residual: 9.9064658446e-07
```

IPython Console Plots Variable Explorer Help Debugger History Files

Dissipated Energy from Earthquake= 168051.84 N·m

Dissipated Energy from Cyclic Displacement= 840276.69 N·m

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

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Seismic: 60

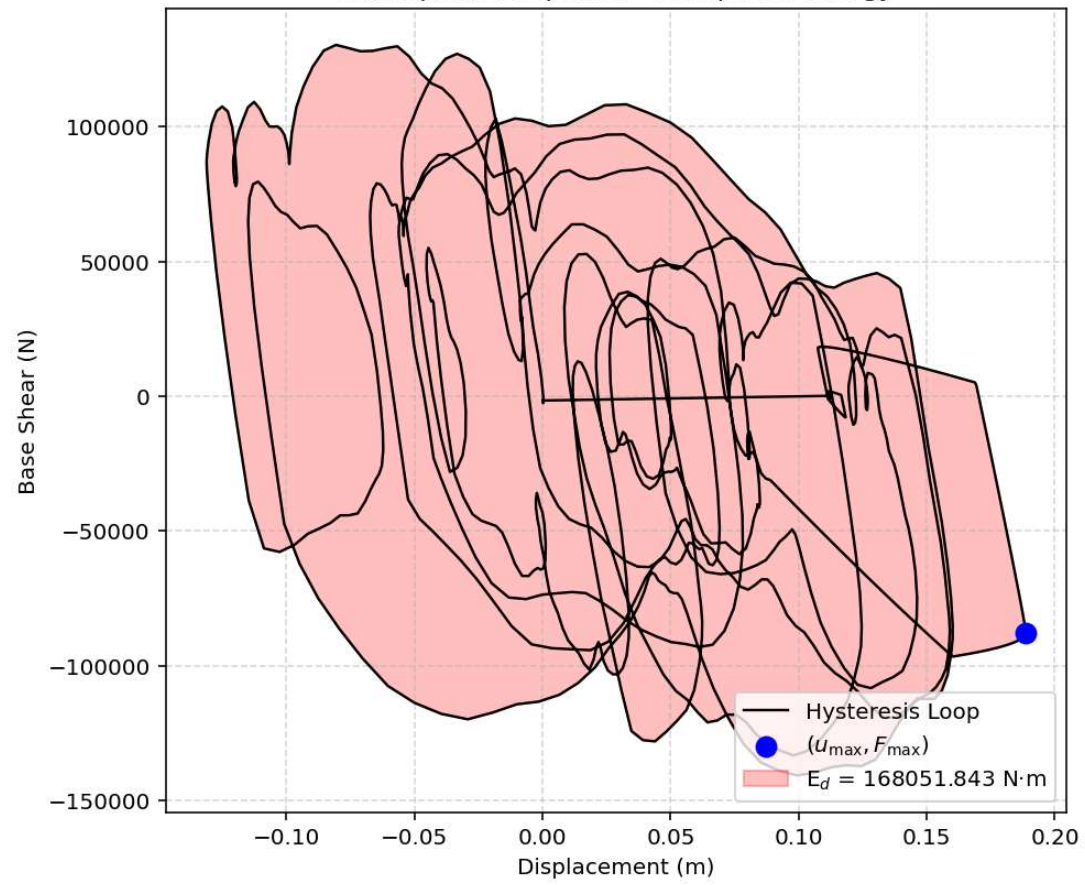
Optimum X (STRU. DUCTILITY RATIO): 49.9470

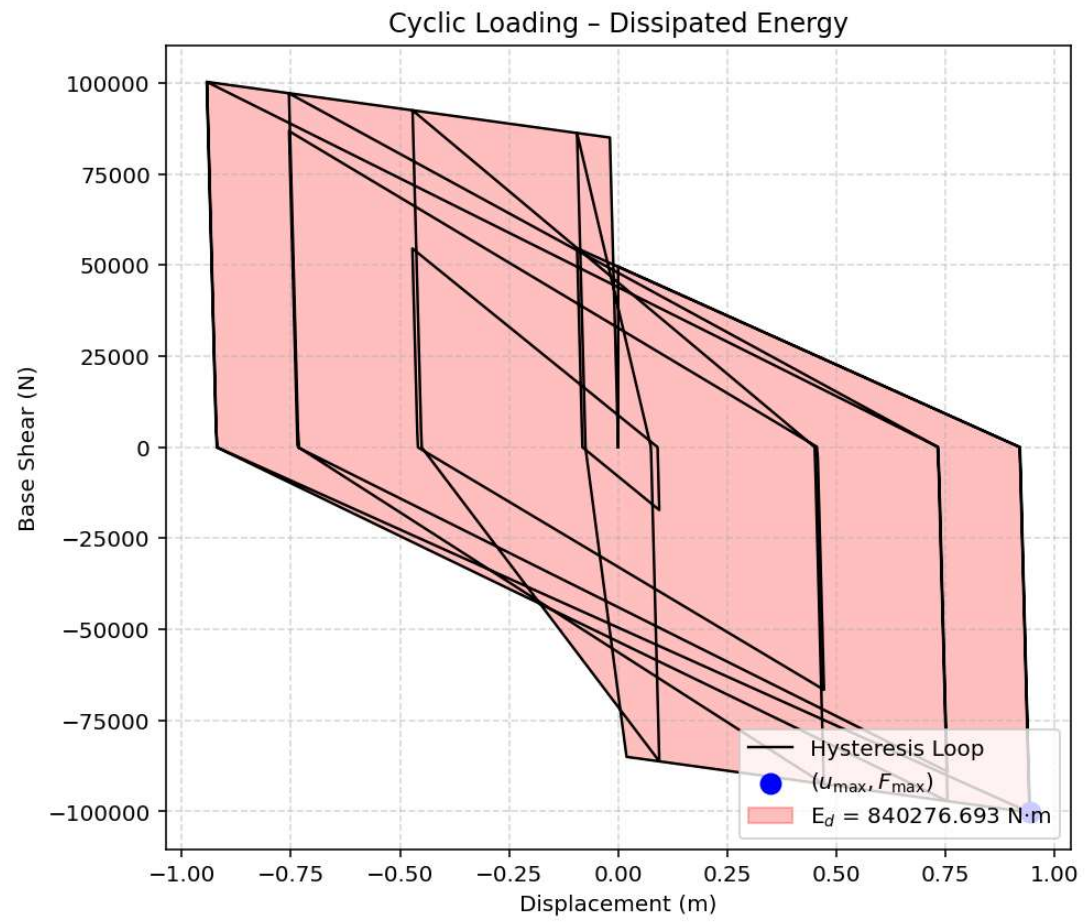
Iteration Counts: 102

Convergence Residual: 9.9064658446e-07

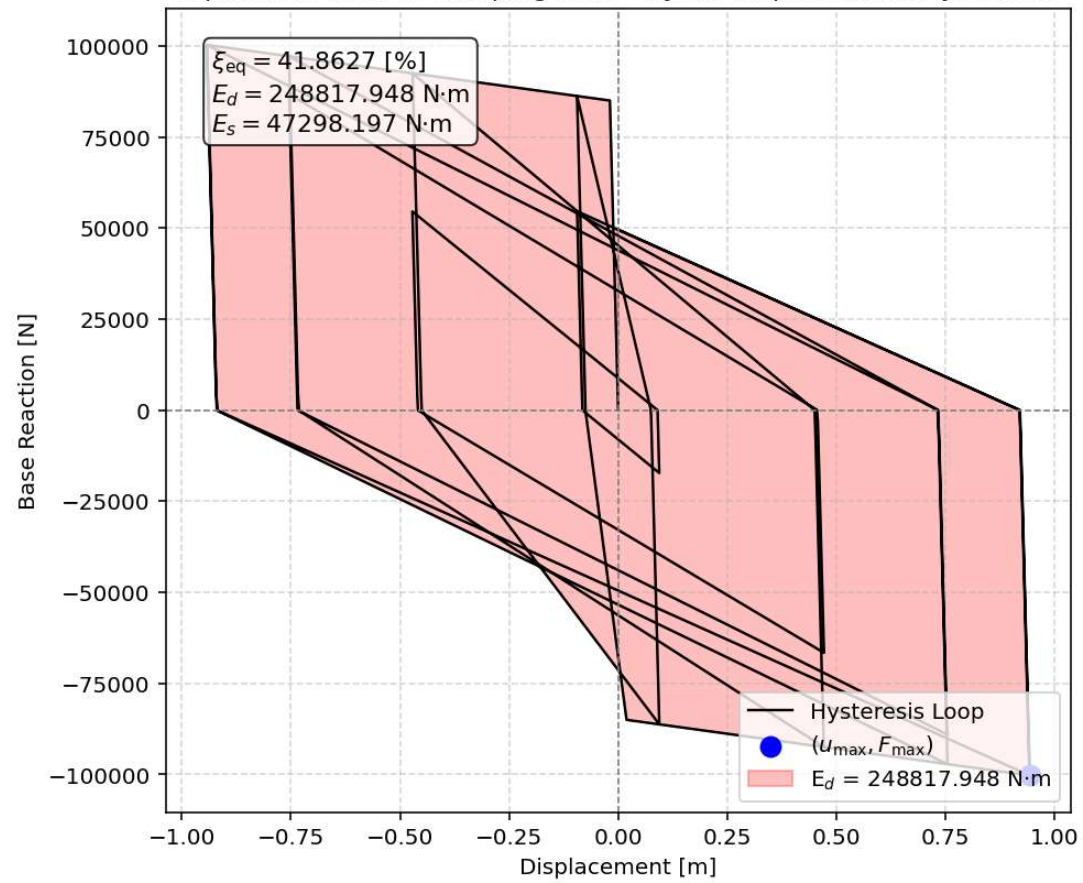
Total time (s): 1103.2188

Earthquake Response - Dissipated Energy

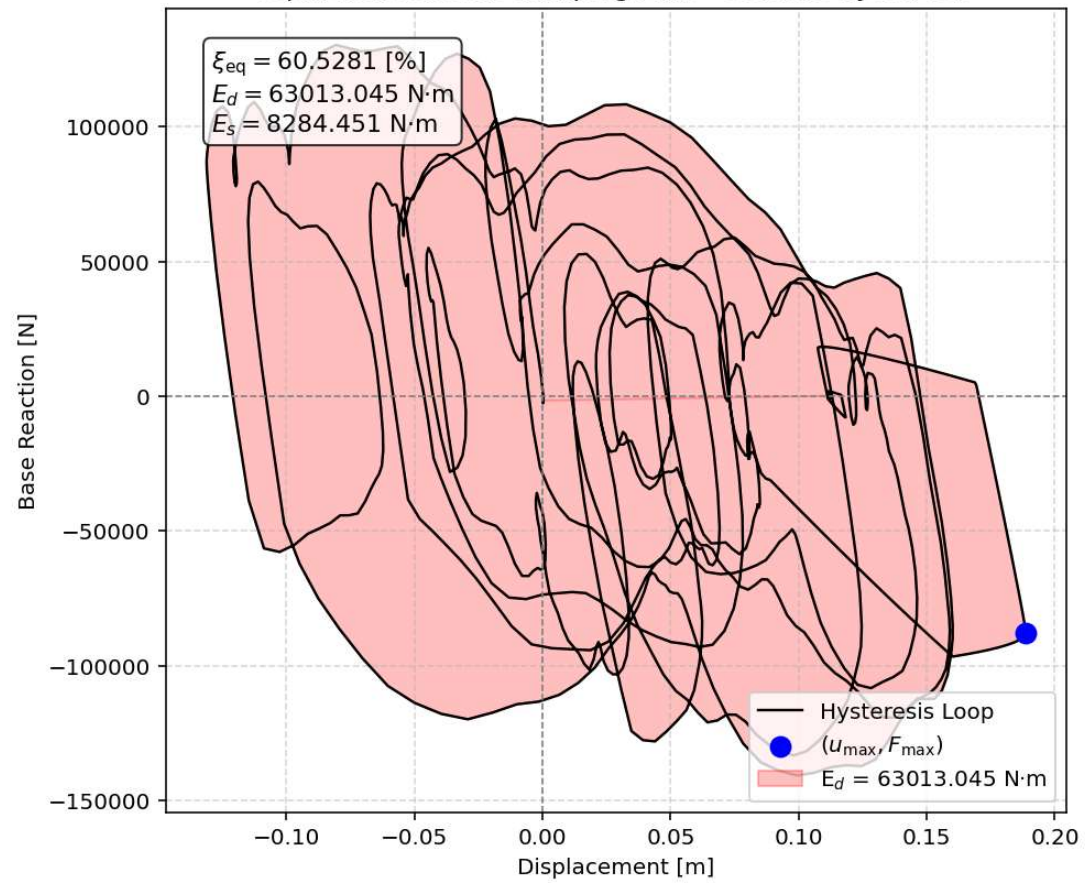




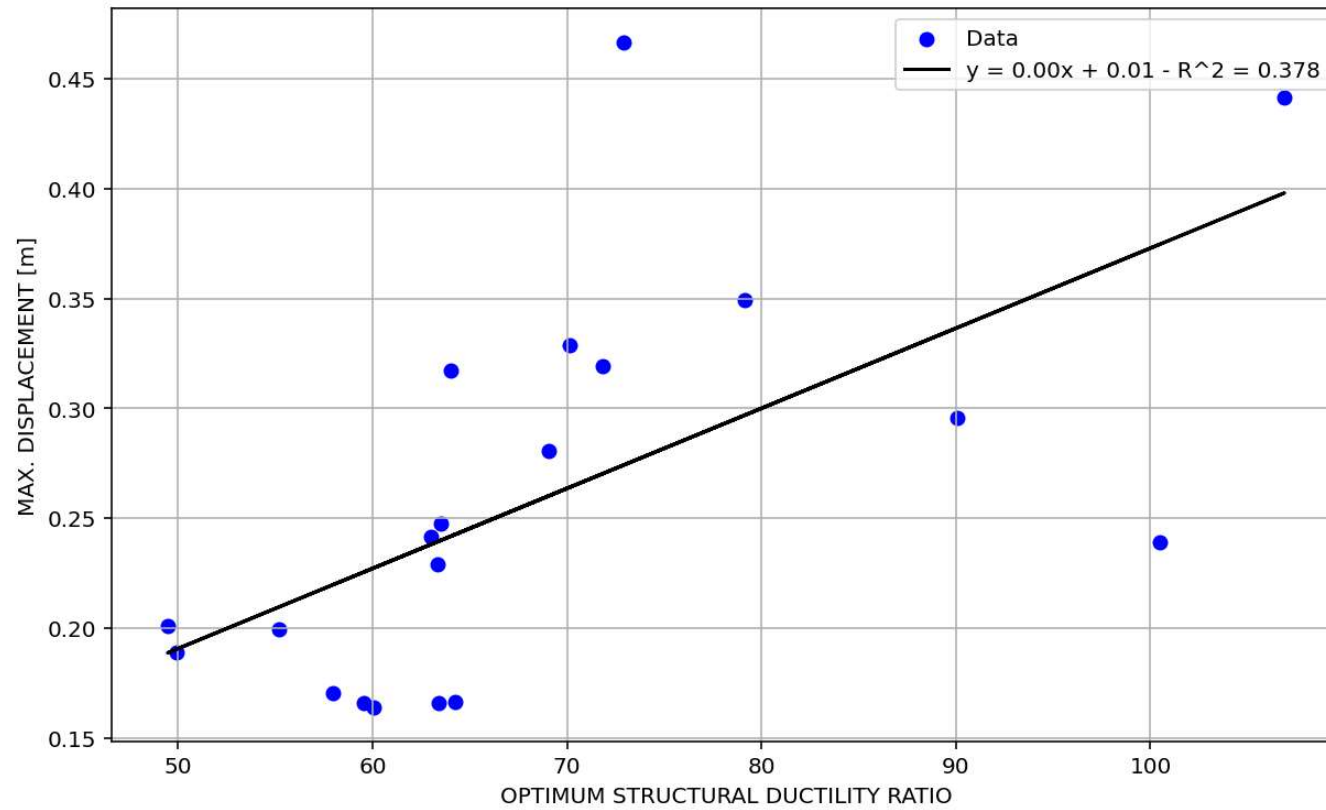
Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



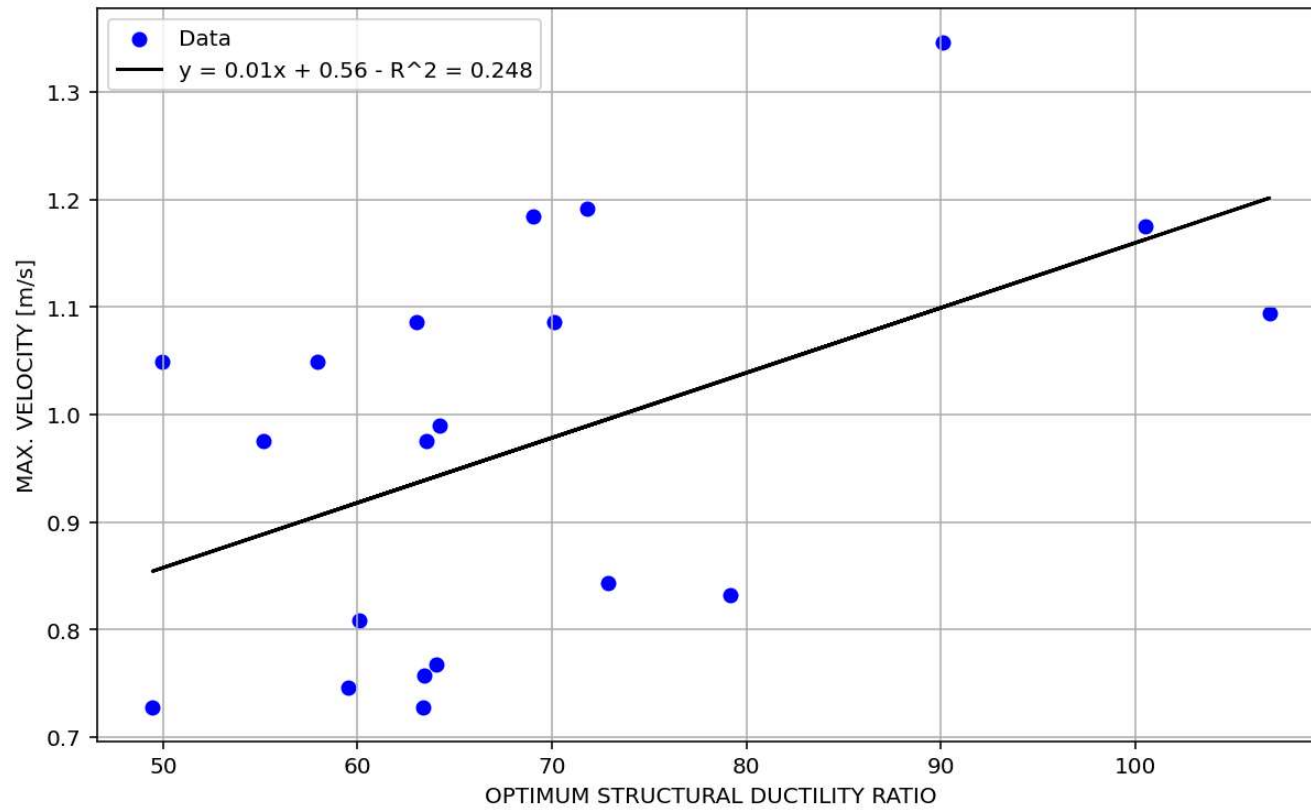
Equivalent viscous damping ratio - Seismic Hysteresis



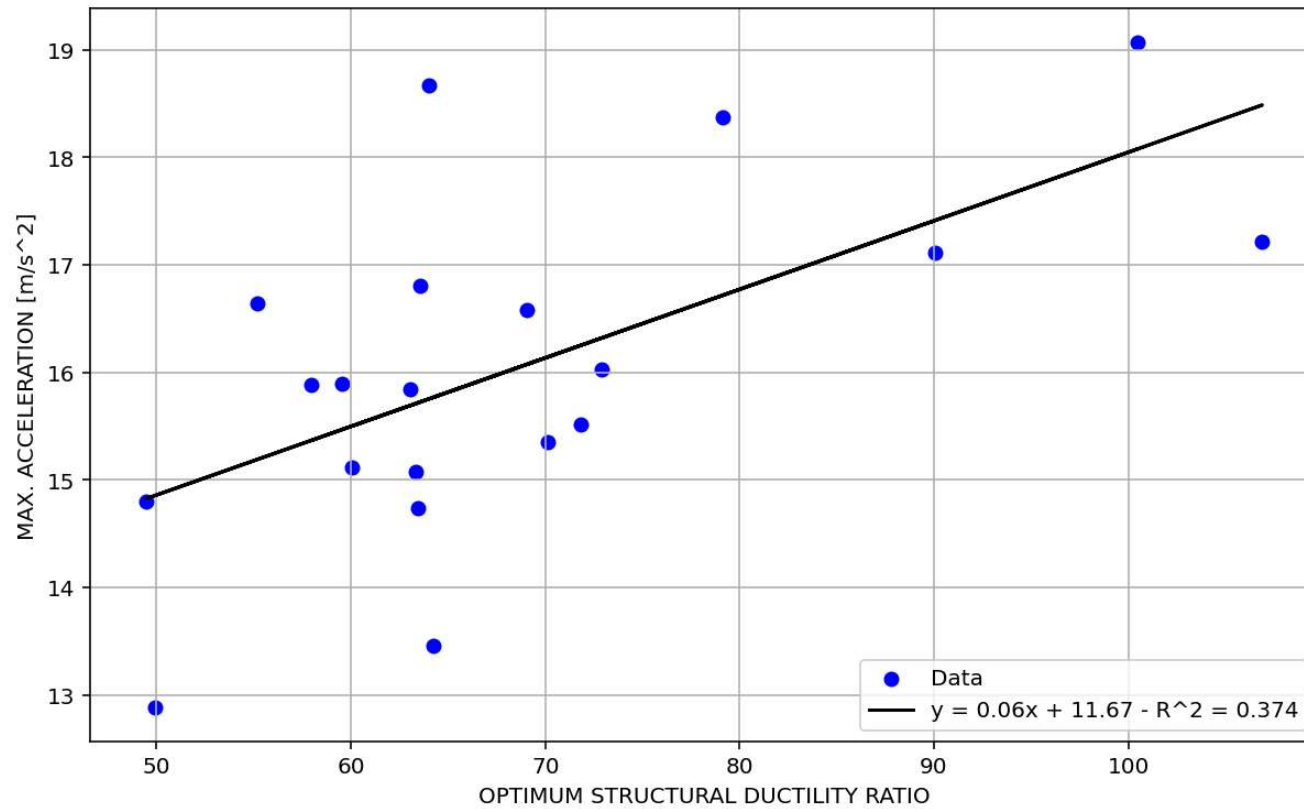
MAX. DISPLACEMENT [m] and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



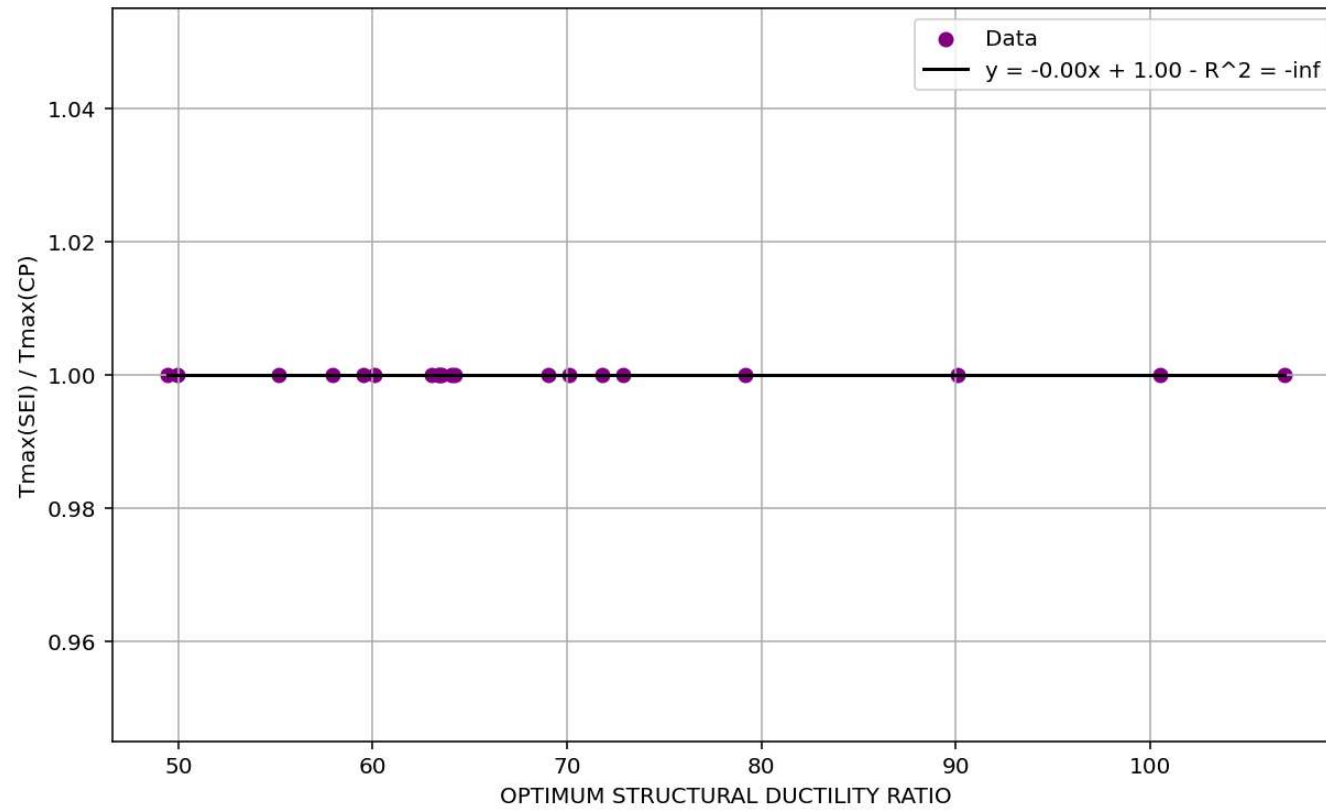
MAX. VELOCITY [m/s] and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



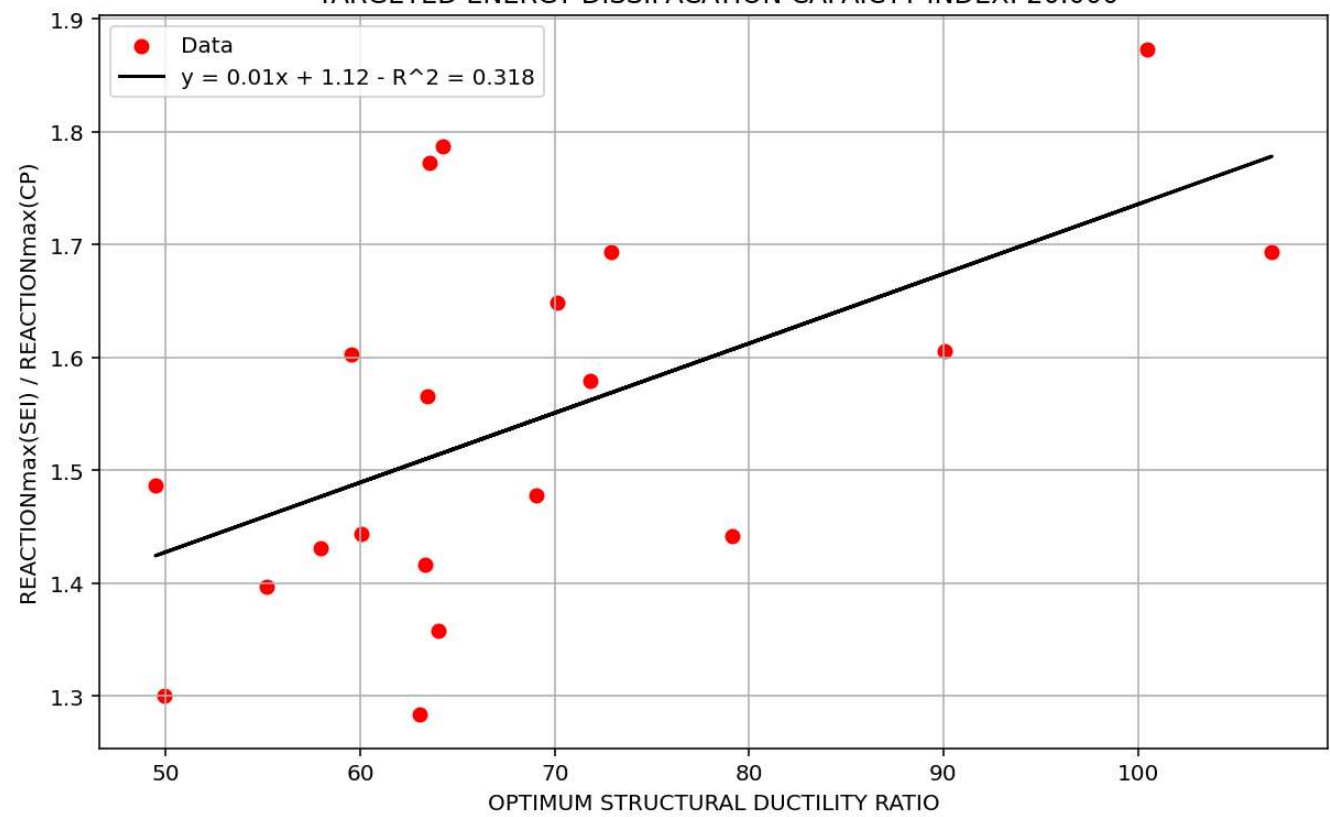
MAX. ACCELERATION [m/s²] and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



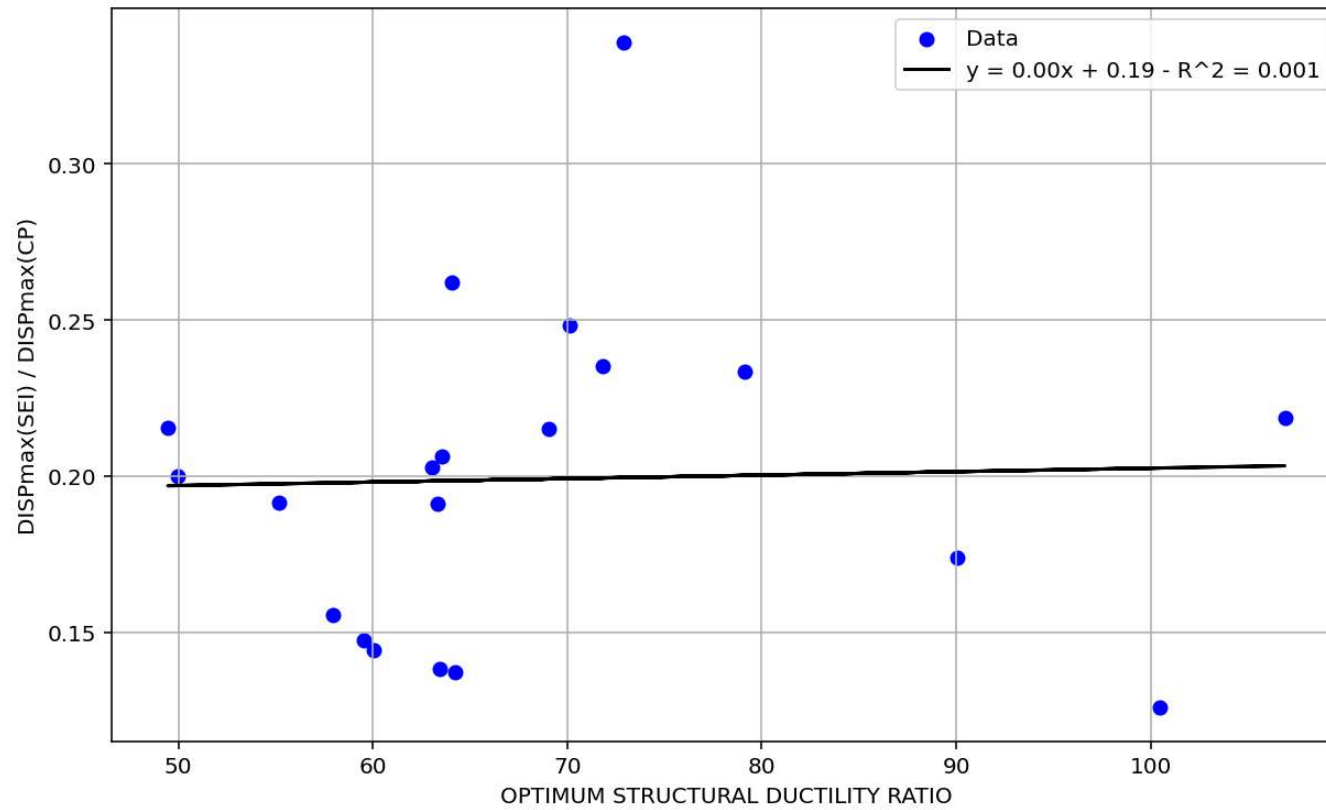
Tmax(SEI) / Tmax(CP) and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



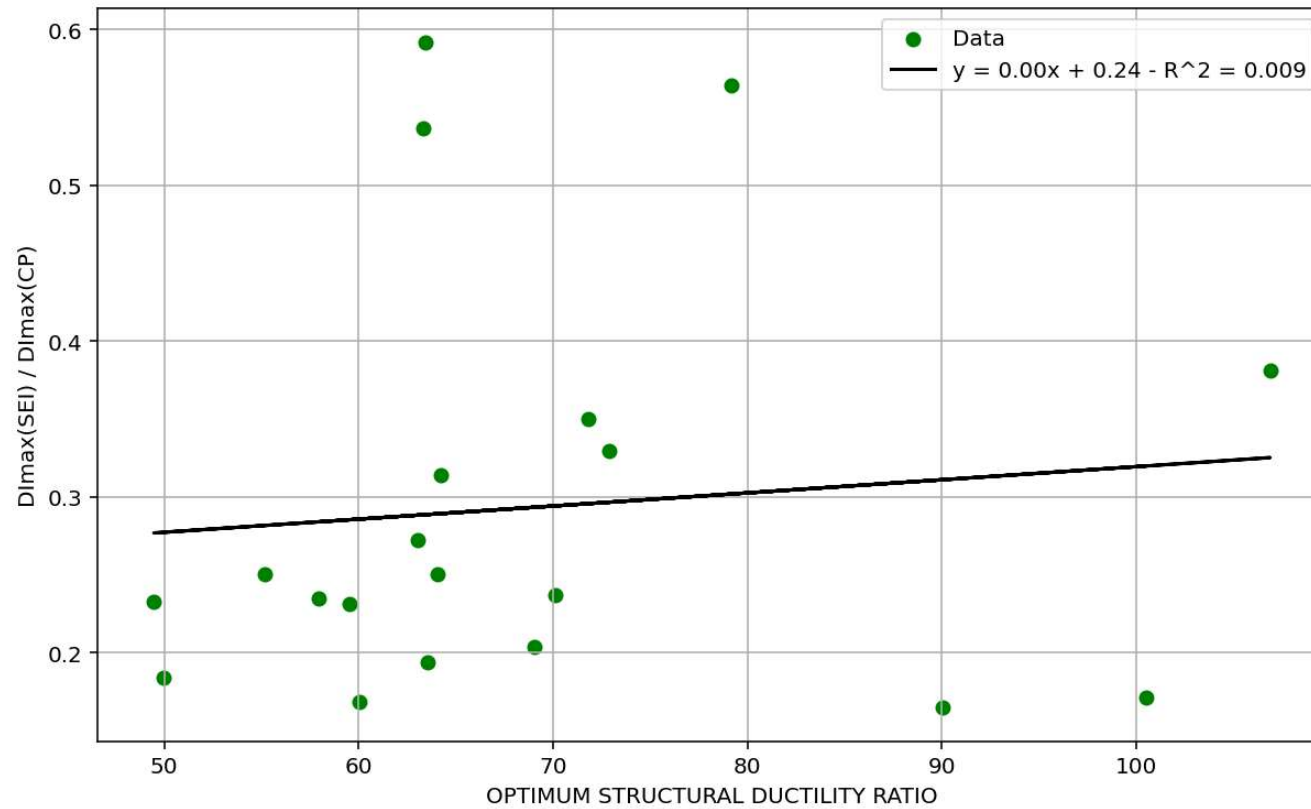
REACTIONmax(SEI) / REACTIONmax(CP) and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



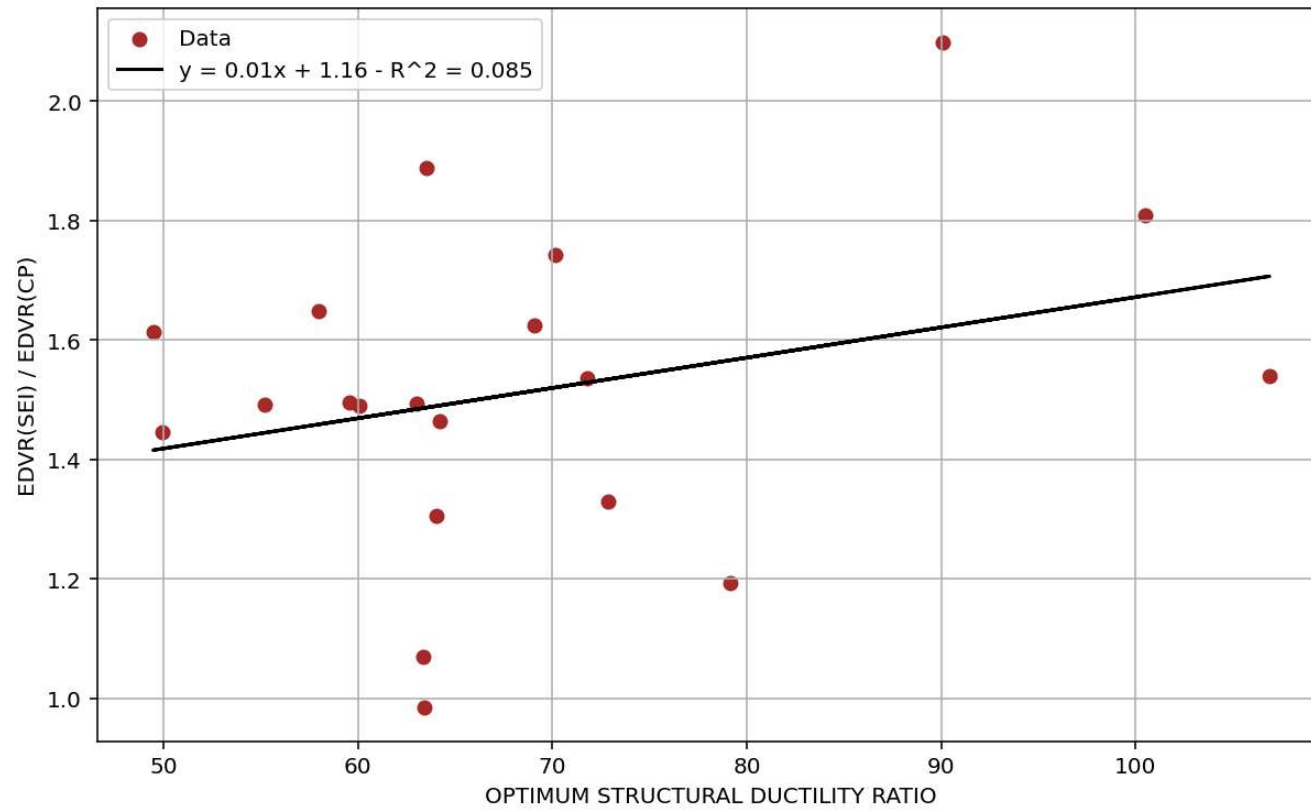
DISPmax(SEI) / DISPmax(CP) and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



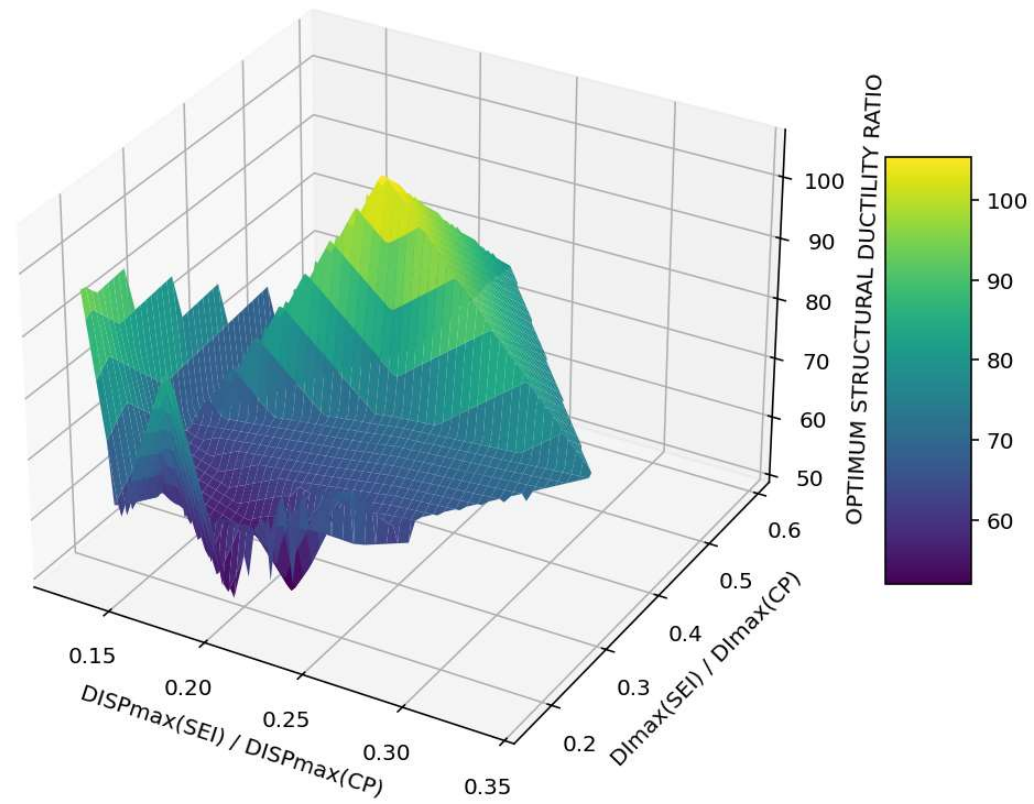
Dlmax(SEI) / Dlmax(CP) and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



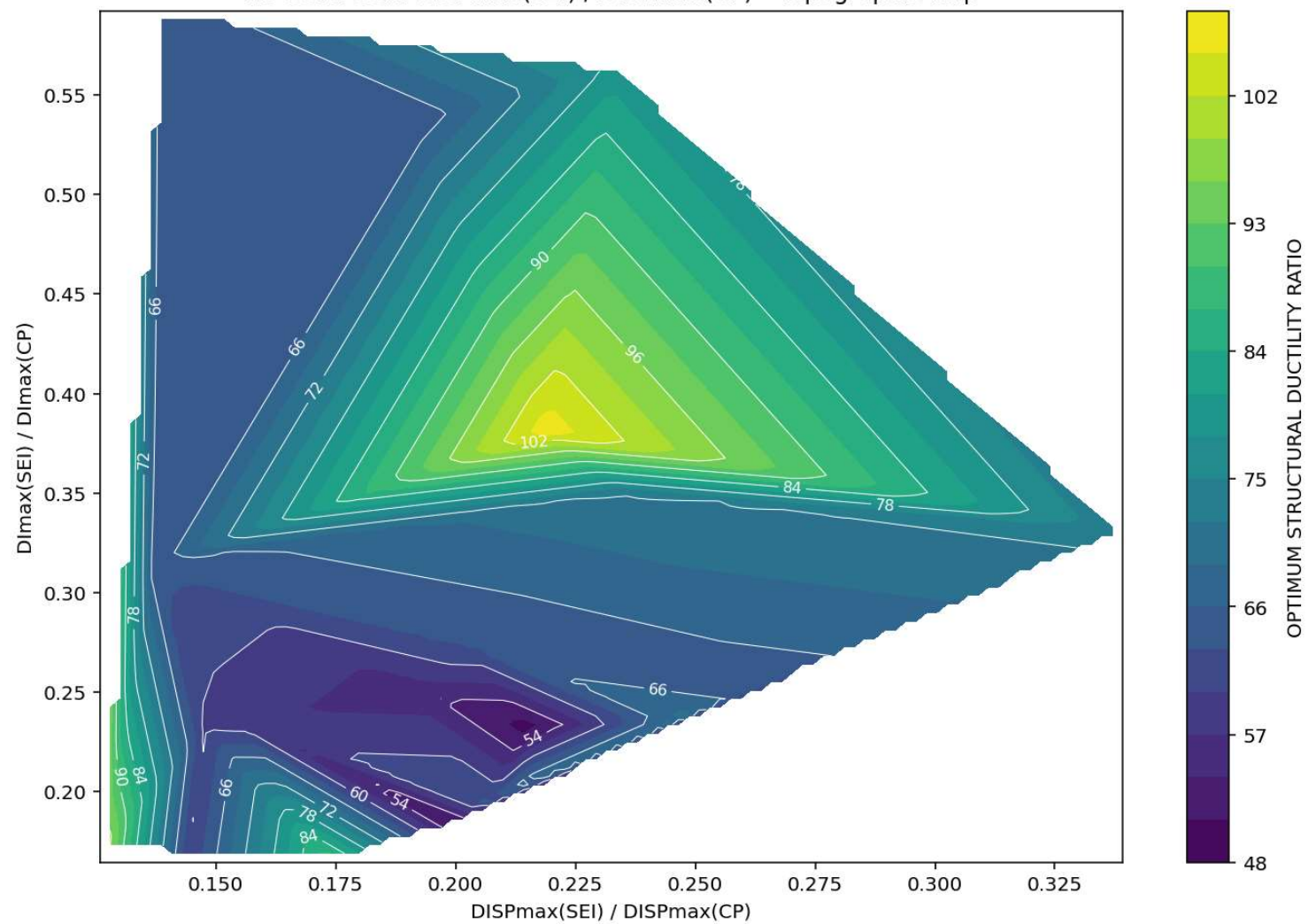
EDVR(SEI) / EDVR(CP) and OPTIMUM STRUCTURAL DUCTILITY RATIO DURING PERIOD ANALYSIS WITH 20 GROUND MOTIONS
TARGETED ENERGY DISSIPACATION CAPAICTY INDEX: 20.000



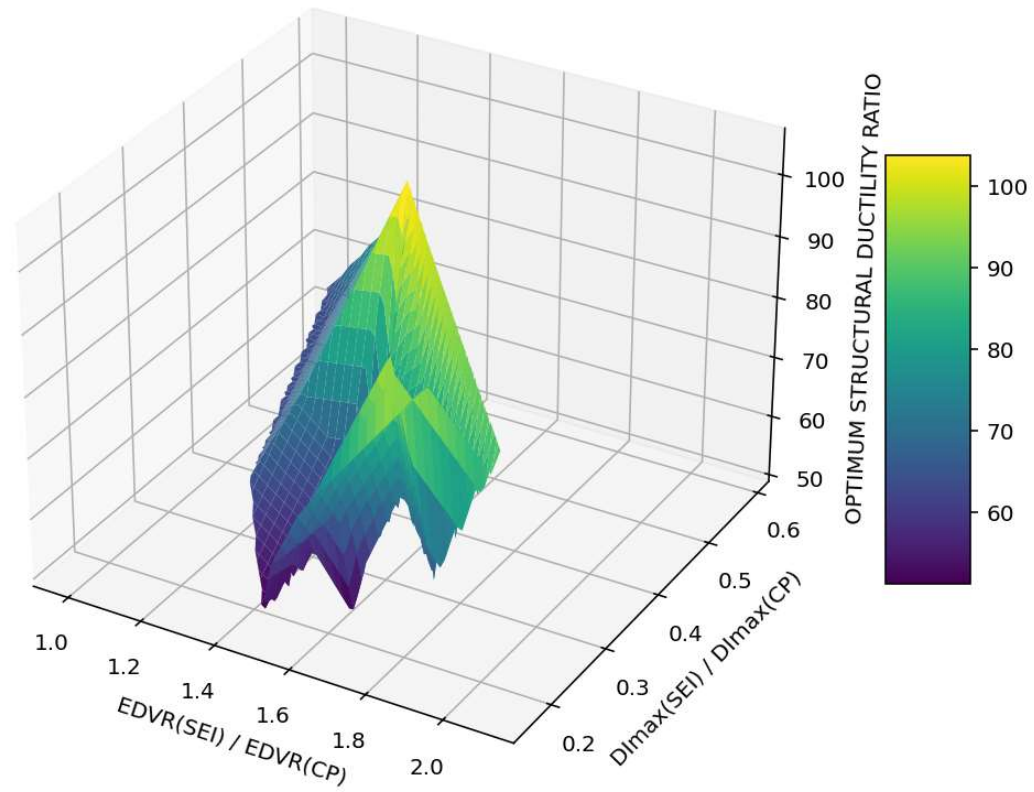
3D CONTOUR: $\text{DISPmax(SEI)} / \text{DISPmax(CP)} - \text{DImax(SEI)} / \text{DImax(CP)} - \text{OPTIMUM STRUCTURAL DUCTILITY RATIO}$

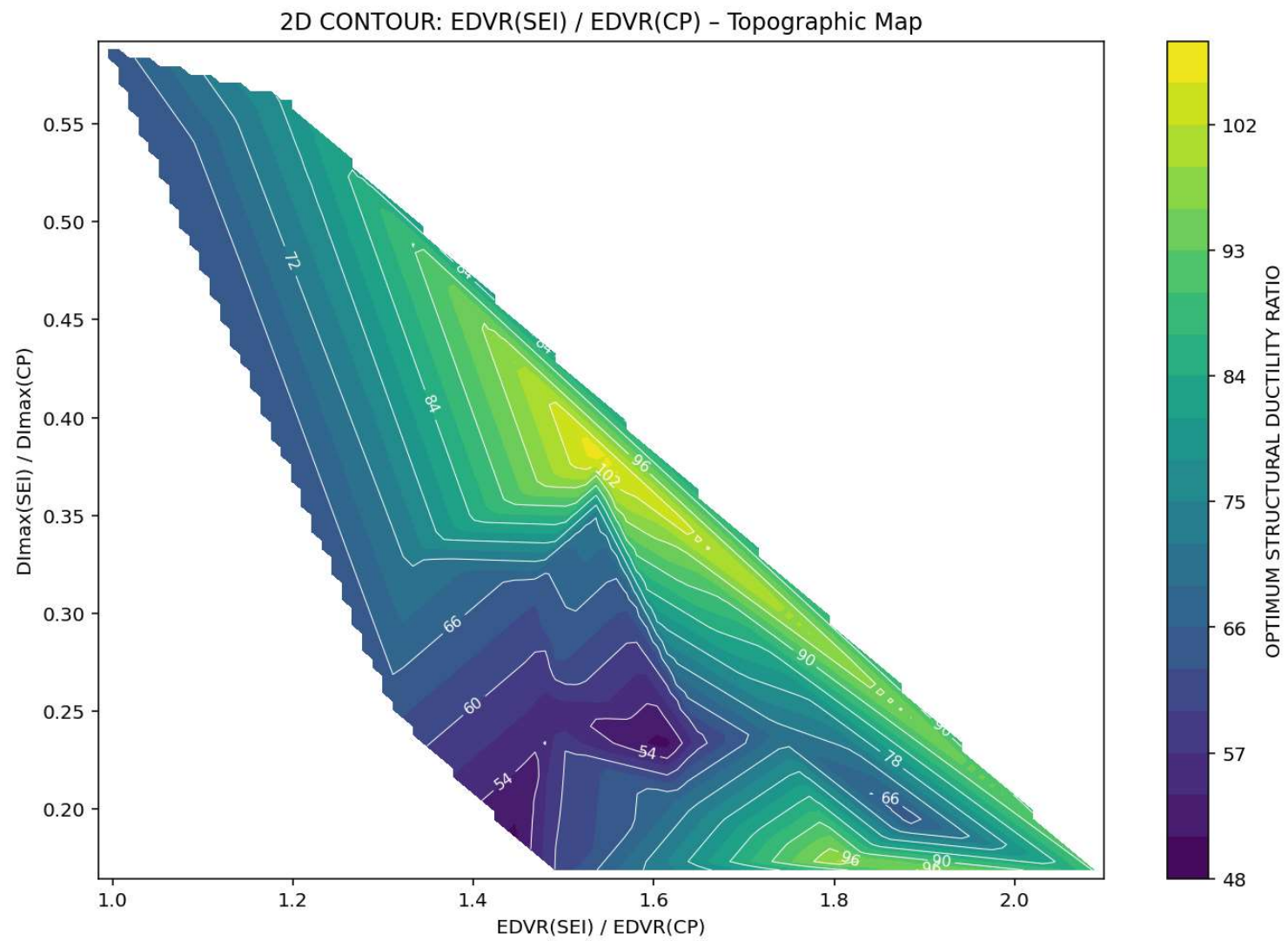


2D CONTOUR: $\text{DISPmax(SEI)} / \text{DISPmax(CP)}$ - Topographic Map

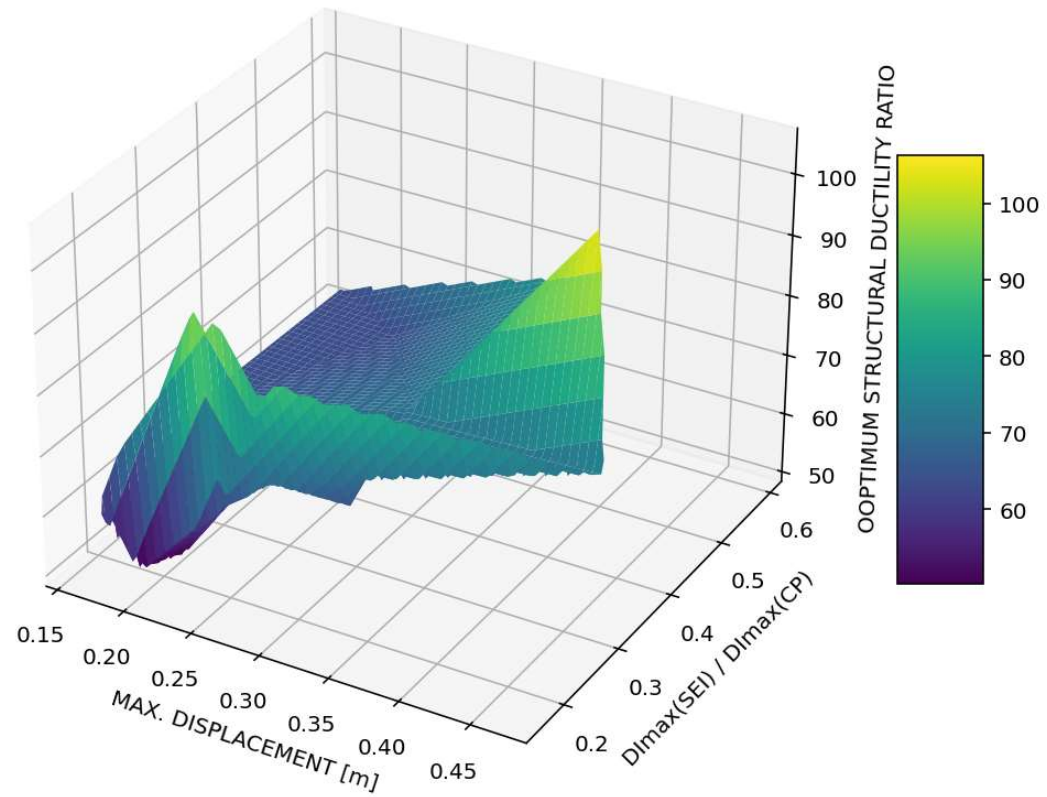


3D CONTOUR: $\text{EDVR}(\text{SEI}) / \text{EDVR}(\text{CP}) - \text{D}_{\text{max}}(\text{SEI}) / \text{D}_{\text{max}}(\text{CP}) - \text{OPTIMUM STRUCTURAL DUCTILITY RATIO}$

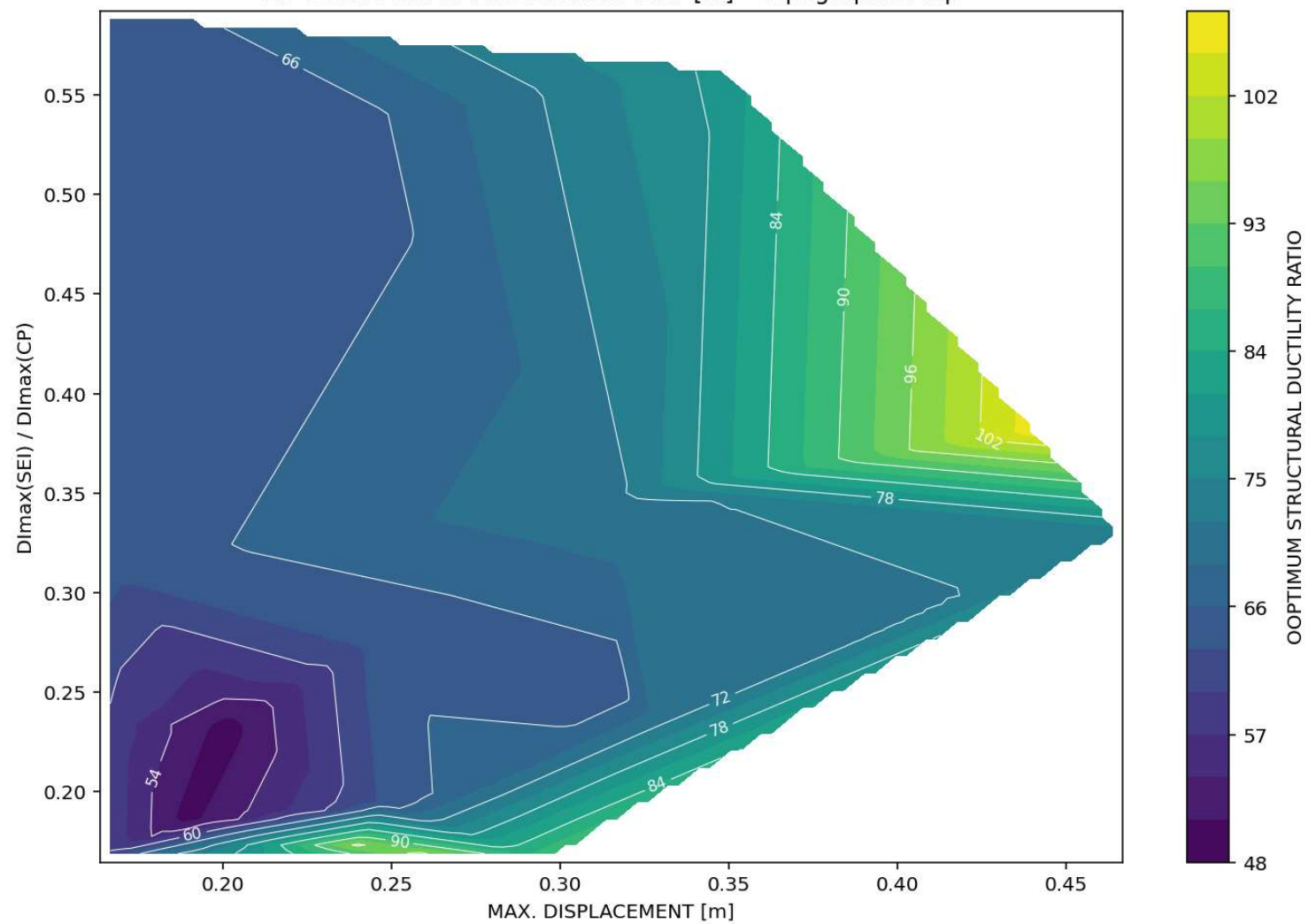




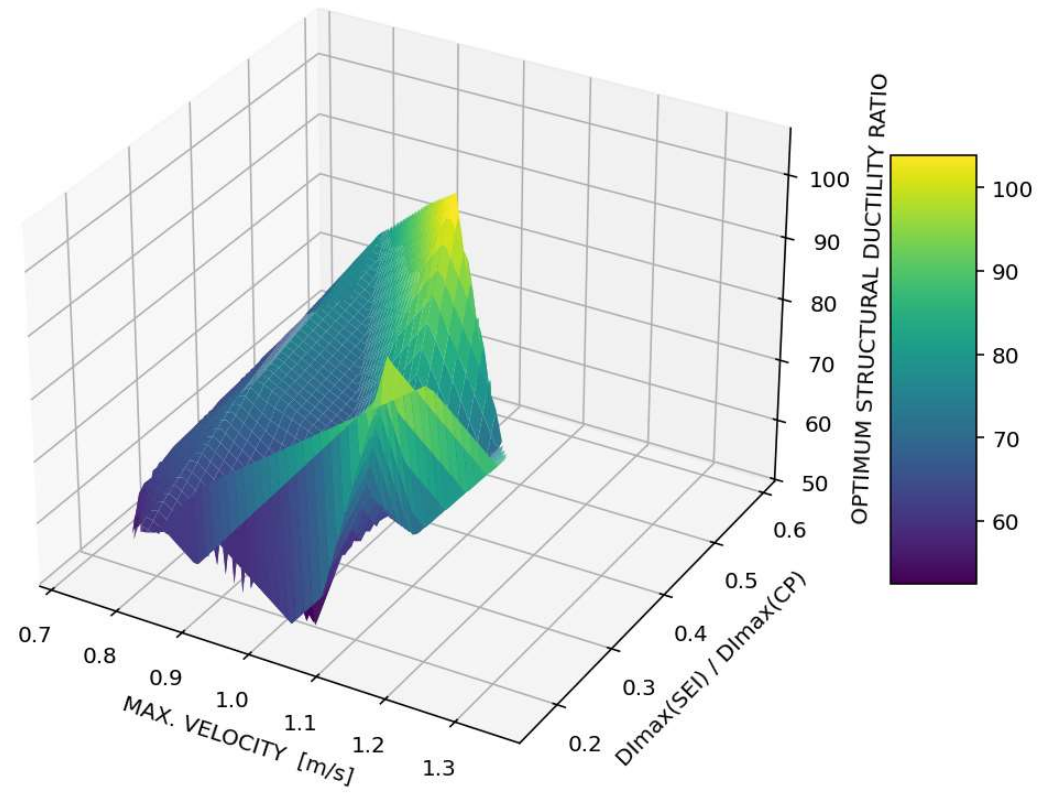
3D CONTOUR: MAX. DISPLACEMENT [m] - $D_{\max}(\text{SEI}) / D_{\max}(\text{CP})$ - OPTIMUM STRUCTURAL DUCTILITY RATIO

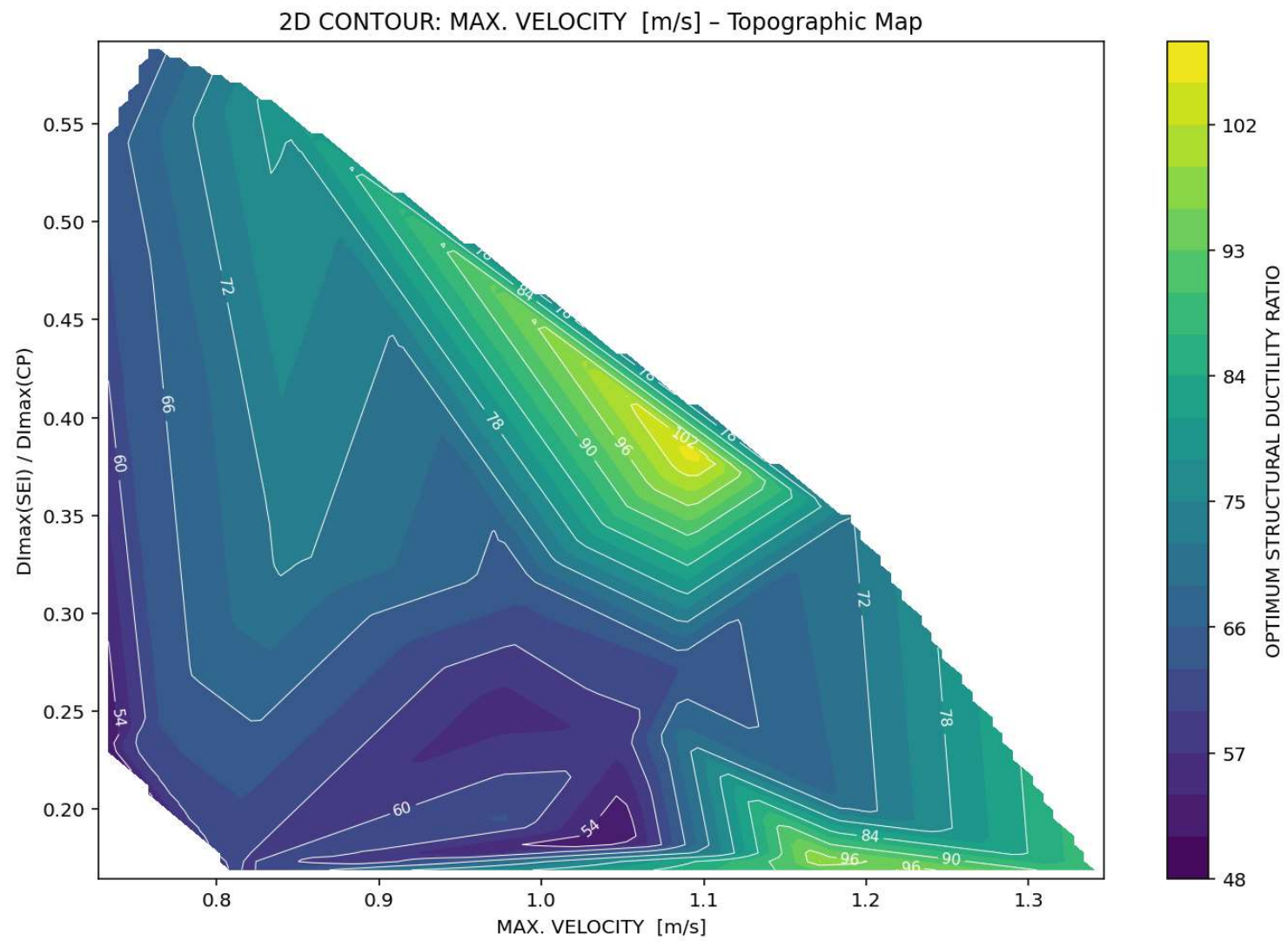


2D CONTOUR: MAX. DISPLACEMENT [m] - Topographic Map

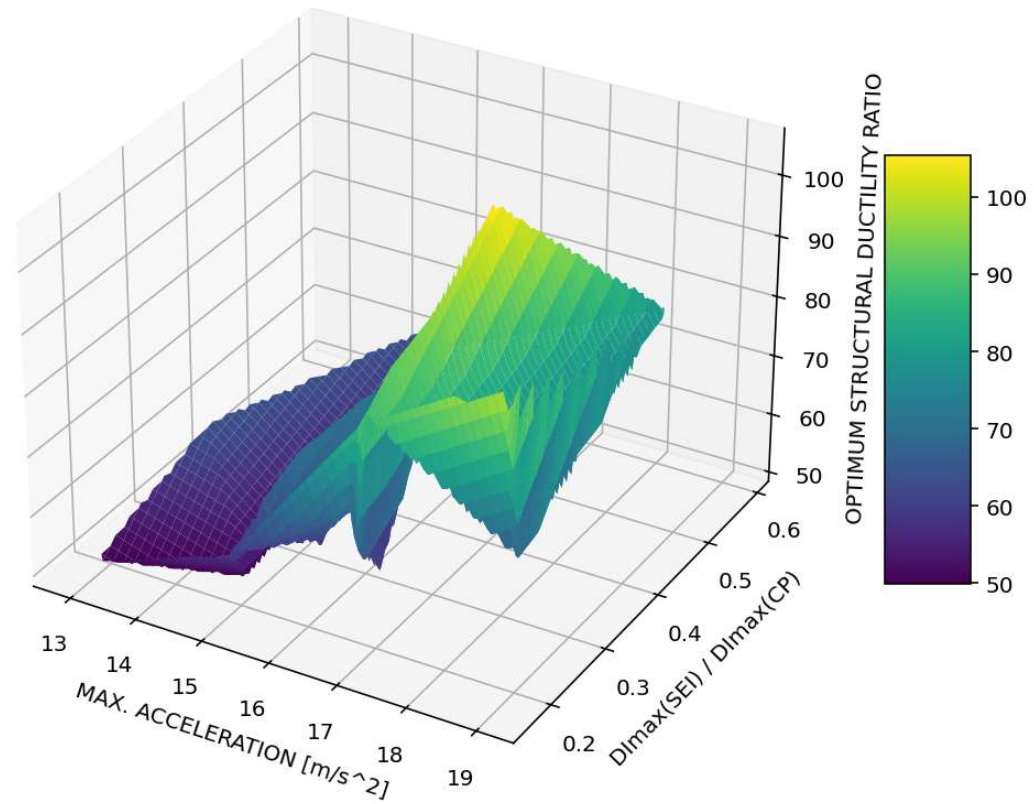


3D CONTOUR: MAX. VELOCITY [m/s] - $D_{\max}(\text{SEI}) / D_{\max}(\text{CP})$ - OPTIMUM STRUCTURAL DUCTILITY RATIO





3D CONTOUR: MAX. ACCELERATION [m/s^2] - $D_{\text{max}}(\text{SEI}) / D_{\text{max}}(\text{CP})$ - OPTIMUM STRUCTURAL DUCTILITY RATIO



2D CONTOUR: MAX. ACCELERATION [m/s²] - Topographic Map

